Practice workbook · Connecting to the internet: switch, WAP, router and modem

Placing and linking internet-access components in a diagram, wired or wireless, in the correct order towards the ISP/internet.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-13

1. Question 1 · part d · 3 marks

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Worked-example retry · Attempt part d only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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1. Question 1 · part d · 3 marks

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Worked-example retry · Attempt part d only.

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(d) A library sets up a network using a star topology.

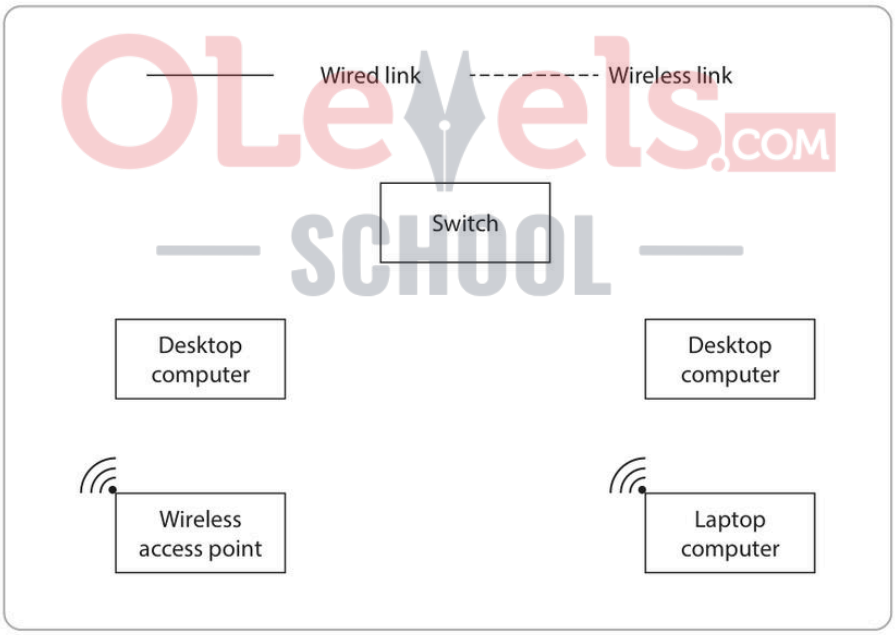
The library is connecting:

- two desktop computers with wired network interface cards
- a laptop with a wireless network interface card
- a switch
- a wireless access point.

Complete the diagram to show how the devices are connected.

Use a solid line to indicate a wired link and a dashed line to indicate a wireless link.

(3)



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1. Question 1 · part d · 3 marks

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Worked-example retry · Attempt part d only.

(e) IT teams carry out network penetration testing.

Describe the purpose of penetration testing.

(2)

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(Total for Question 1 = 12 marks)

Working and answer

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2. Question 2 · part d · 6 marks

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Further practice · Attempt part d only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

.....

.....

Cloud storage provider

.....

.....

(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	**Zafer**
Washing machine design drawings		
A folder of manuals for new machines		

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2. Question 2 · part d · 6 marks

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Further practice · Attempt part d only.

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(c) Zafer uses a browser to view pages on Pearson's website by typing a uniform resource locator into a browser. This is shown in **Figure 1**.



Figure 1

Pearson's machine	Zafer's machine
MAC address: 03-57-00-EC-4B-30	MAC address: 00-15-00-BC-9A-90
IPv4 address: 2.20.38.113	IPv4 address: 192.168.1.78
IPv4 Subnet Mask: 255.255.0.0	IPv4 Subnet Mask: 255.255.255.0
Website domain: www.pearson.com	Web browser
Main web page: https://www.pearson.com/uk/	Uniform resource locator: www.pearson.com

A domain name server is used in this process.

Identify the input to and the output from the domain name server.

(2)

Input

Output

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2. Question 2 · part d · 6 marks

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Further practice · Attempt part d only.

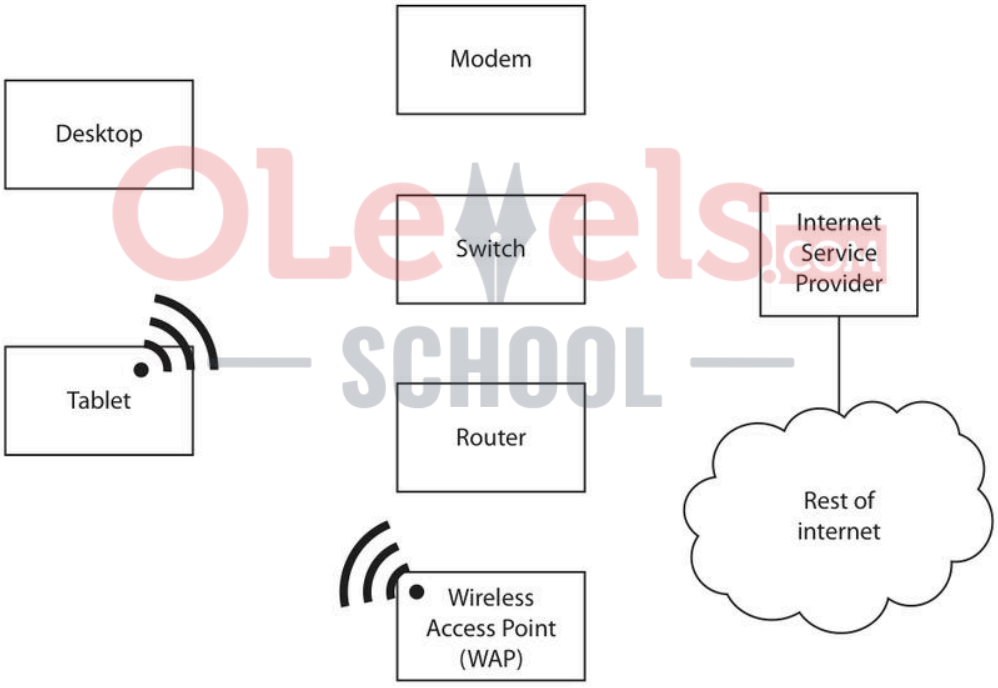
(d) Zafer can use a desktop or a tablet computer to connect to the internet.

Only the tablet has wireless capabilities.

The diagram shows the components used to connect to the internet.

Complete the diagram to show how the desktop and the tablet are connected to the internet.

(6)



(Total for Question 2 = 12 marks)

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2. Question 2 · part d · 6 marks · Working and answer

Working and answer

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3. Question 4 · part d · 3 marks

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Further practice · Attempt part d only.

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4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

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(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

3. Question 4 · part d · 3 marks

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Further practice · Attempt part d only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

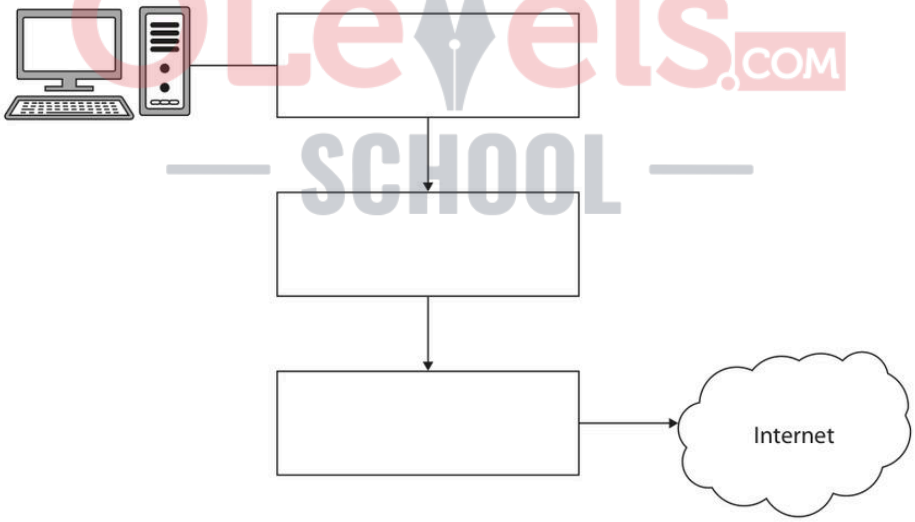
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



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3. Question 4 · part d · 3 marks

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Further practice · Attempt part d only.

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(e) Users enter passwords when logging onto a network and when creating accounts using a web page.

Describe **one** difference between validation and authentication.

(2)

(Total for Question 4 = 12 marks)

Working and answer

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